Notes: Piskorski, Seru, and Vig (JFE, 2010)

Securitization and Distressed Loan Renegotiation: Evidence from the Subprime Mortgage Crisis

Contents

1 Big picture

- **Question.** Does securitization make distressed mortgage loans harder to renegotiate? The paper studies whether servicers foreclose more aggressively when a delinquent loan has been securitized rather than kept on the lender's own balance sheet.
- **Setting.** The empirical setting is the U.S. non-agency mortgage market during the subprime crisis. The authors use loan-level servicing data on first-lien mortgages originated in 2005 and 2006 and track payment outcomes through the first quarter of 2008.
- **Why this matters.** A delinquent loan can be handled in many ways: formal modification, forbearance, repayment plan, short sale, foreclosure moratorium, refinancing into a new loan, transfer to a specialized servicer, or simply waiting rather than foreclosing immediately. If securitization distorts those choices, it can amplify a foreclosure crisis.
- **Main mechanism.** The paper argues that securitization can create a *foreclosure bias*. Possible reasons include weaker servicer incentives, contractual restrictions in pooling and servicing agreements, coordination problems among dispersed investors, and differences in accounting or regulatory treatment between securitized and portfolio loans.
- **Headline result.** Conditional on a loan becoming seriously delinquent, bank-held loans are much less likely to be foreclosed than comparable securitized loans. Across baseline specifications and origination vintages, the foreclosure rate of delinquent bank-held loans is about 3 to 7 percentage points lower in absolute terms, or roughly 13% to 32% lower in relative terms.
- **Cross-sectional pattern.** The effect is small for the lowest-credit-quality borrowers and much larger for medium- and high-quality borrowers. That pattern is important because it is hard to square with a simple "securitized loans are just worse" story.
- **Quasi-experimental confirmation.** The paper exploits repurchase clauses and early payment default clauses around a 90-day securitization threshold. Loans that become delinquent just before the threshold can be pushed back to the originator and then serviced like portfolio loans. Relative to comparable loans becoming delinquent just after the threshold, these repurchased loans are foreclosed about 6.5 percentage points less often.
- **What the paper does *not* fully pin down.** The paper is largely agnostic about the exact channel and about welfare. It shows that securitization changes servicing outcomes; it does not fully prove whether banks are always making the ex post efficient decision.
- **Economic takeaway.** The paper's central message is that securitization is not just a funding choice. It also changes the governance of distress. During the foreclosure crisis, that governance difference mattered a lot.

2 Data and measurement (key objects)

2.1 Main data source

The core dataset comes from Lender Processing Services (LPS, formerly McDash Analytics), a loan-level mortgage servicing database. It contains detailed origination information, geography, payment history, and an indicator for whether the loan is securitized to private investors or held on the lender's balance sheet.

Key origination variables include:

- FICO score,
- documentation level,
- loan-to-value ratio (LTV),
- original interest rate,
- original loan amount,
- maturity,
- fixed-rate versus adjustable-rate status,
- insurance status,
- MSA location,
- investor type.

2.2 Sample construction

The authors restrict attention to:

- first-lien non-agency mortgages,
- originated in 2005 and 2006,
- in continental U.S. locations,
- with maturities of 15, 20, or 30 years,
- with FICO between 500 and 850,
- with LTV below 150,
- and with loans entering the LPS database within four months of origination.

The loans are followed until the end of 2008Q1. The authors stop there because the institutional environment changed after major policy interventions in 2008 and 2009.

2.3 Scale of the dataset

After filtering, the sample contains about 6.2 million unique mortgages. Of these, about 327,000 become 60+ days delinquent. Roughly 75% of loans appear as portfolio loans at origination, but most are securitized within a few months, so only about 11.3% of delinquent loans are bank-held at the time they first become seriously delinquent.

Interpretation. This is a very large dataset for studying servicing outcomes, and the fact that it records ownership status at delinquency is crucial for the paper's question.

2.4 Outcome variables

Serious delinquency. The main regressions condition on a loan becoming 60+ days delinquent under the Mortgage Bankers Association definition.

Foreclosure. A loan is classified as foreclosed if it enters foreclosure post-sale status or REO status in the observed payment history.

Cure. In the cure-rate analysis, a delinquent loan is considered cured if its payment status improves to better than 60+ delinquent within the horizon the authors study.

Transfer. Some delinquent loans disappear from LPS because servicing rights move to servicers outside LPS coverage. This motivates the hazard-model analysis.

2.5 Key explanatory variable

The central explanatory variable is:

$$\text{Portfolio}_i = \begin{cases} 1 & \text{if the delinquent loan is bank-held at first 60+ delinquency,} \\ 0 & \text{if the delinquent loan is privately securitized.} \end{cases}$$

The empirical question is whether $\text{Portfolio}_i = 1$ predicts a lower foreclosure probability for otherwise similar delinquent loans.

2.6 Borrower quality measures

The paper emphasizes two forms of ex ante borrower quality:

- **FICO score,** the standard hard-information summary of credit risk.
- **Documentation status,** split into full, limited, and no documentation.

A particularly important subsample is the "high-quality" group:

- full documentation, and
- FICO at least 680.

This subsample contains about 1.3 million loans, around 16,500 of which become seriously delinquent. About 20.4% of those delinquent high-quality loans are bank-held at delinquency.

2.7 Why the high-quality sample matters

The authors argue that screening on unobservables should be less important for borrowers who look strong on hard information. If differences between portfolio and securitized loans remain large in that group, the simple "banks keep the good loans" story becomes less convincing.

2.8 Special sample for the quasi-experiment

In Section 6, the paper builds a special sample around repurchase clauses:

- loans that first become 30+ delinquent near 90 days after securitization,
- then transition to 60+ delinquency in the next month,
- and are either repurchased by the originator or remain securitized.

That sample is the core of the paper's causal design.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline foreclosure specification

The main reduced-form specification is a logit model estimated on delinquent loans:

$$\Pr(\text{Foreclosure}_i = 1 \mid \text{Delinquent}) = F\left(\alpha + \beta\,\text{Portfolio}_i + \gamma' X_i + \delta_m + \varepsilon_i\right), \tag{1}$$

where:

- Foreclosure_i indicates whether the delinquent loan is eventually foreclosed,
- Portfolio_i indicates whether the loan is bank-held at the time of first 60+ delinquency,
- X_i contains borrower and loan observables,
- δ_m are MSA fixed effects,
- and $F(\cdot)$ is the logit link.

Interpretation of β. A negative β means bank-held delinquent loans are less likely to be foreclosed. Equivalently, securitized delinquent loans are foreclosed more aggressively.

3.2 Why the regression conditions on delinquency

The paper is not studying whether securitized loans are more likely to *become* delinquent. It is studying what happens *after* serious delinquency occurs. Conditioning on 60+ delinquency focuses the comparison on the servicer's workout-versus-foreclosure decision.

3.3 Controls in the baseline model

The baseline regression is intentionally rich. Controls include:

- FICO-category dummies,
- LTV and LTV squared,
- original loan amount and its square,
- original interest rate,
- fixed-rate dummy,

- 15-year and 20-year term dummies,
- insurance indicators,
- age at delinquency,
- MSA fixed effects.

The authors also estimate regressions separately by origination quarter to avoid pooling together loans born in different macro environments.

3.4 Using information at delinquency

A potential concern is that lenders learn more about borrower quality between origination and delinquency. To address this, the authors re-estimate the portfolio effect using updated FICO and updated LTV measured at delinquency for the subsample where those data are available.

Interpretation. If the main result were just about new information revealed after origination, the portfolio effect should collapse once those updated measures are included. It does not.

3.5 Hazard model for foreclosure and transfer

Because some delinquent loans are transferred out of LPS coverage, the paper also estimates a Cox proportional hazard model that allows three states: foreclosed, not foreclosed, or transferred away from the observable sample.

A convenient summary form is:

$$h_i(t) = h_0(t) \exp\big(\beta \, \mathrm{Portfolio}_i + \gamma' X_i + \tau_t + \delta_m\big) \,, \tag{2}$$

where $h_i(t)$ is the hazard of transition into foreclosure (or transfer) at time t.

Interpretation. This lets the authors use the available payment history for transferred loans rather than dropping them mechanically.

3.6 Cure-rate specification

The authors also estimate a hazard model where the event is a cure rather than a foreclosure. This asks whether delinquent borrowers with bank-held loans are more likely to resume making payments than comparable borrowers with securitized loans.

Important point. Cure rates and foreclosure rates should be read together. A loan that is not cured is not the same thing as a loan that has already been foreclosed. A lender that waits instead of foreclosing preserves the option to renegotiate later.

3.7 Cross-sectional heterogeneity

To study heterogeneity, the sample is split into FICO bins:

- FICO < 620,

- $620 \leq \text{FICO} < 680$,
- $\text{FICO} > 680$.

The same foreclosure and cure specifications are estimated in each bin.

3.8 Quasi-experimental design around repurchase clauses

The paper's cleanest causal design exploits repurchase and early payment default provisions that can force originators to take back some securitized loans that become delinquent within roughly the first 90 days after securitization.

The core empirical specification is:

$$\Pr(Y_{imt} = 1 \mid \text{Delinquent}) = F\left(\alpha + \tau_t + \beta\, 1\{i \in T\} + \gamma' X_{imt} + \delta_m + \varepsilon_{imt}\right), \tag{3}$$

where:

- Y_{imt} is a foreclosure indicator,
- $1\{i \in T\}$ identifies loans in the treatment group,
- X_{imt} contains borrower and contract controls,
- τ_t are origination-time fixed effects,
- δ_m are MSA fixed effects.

Treatment group in the main 3:4 test.

- The loan becomes 30+ delinquent in month 3 after securitization,
- becomes 60+ delinquent in month 4,
- and is repurchased by the originator within the next three months.

Control group in the main 3:4 test.

- The loan becomes 30+ delinquent in month 4 after securitization,
- becomes 60+ delinquent in month 5,
- and remains securitized in the next three months.

Interpretation of β. If observables and unobservables are smooth around the 90-day threshold, β measures the causal effect of moving a distressed loan from securitized servicing to portfolio-style servicing.

4 Identification and instruments (what makes the estimates credible)

4.1 Main identification problem

The obvious threat is selection: perhaps lenders securitize worse loans and keep better loans on balance sheet. If so, securitized loans would naturally foreclose more often even without any servicing distortion.

4.2 How the paper addresses selection in the baseline analysis

The paper responds in several ways:

- It conditions on a loan already becoming seriously delinquent, so the comparison is among distressed loans rather than among all originated loans.
- It uses rich origination controls, MSA fixed effects, and flexible nonlinear terms.
- It re-runs the analysis on a high-quality subsample where screening on soft information should be less important.
- It adds updated FICO and LTV at delinquency to absorb information revealed between origination and delinquency.

Interpretation. None of these steps is perfect on its own, but together they make the "pure selection" explanation substantially less persuasive.

4.3 Attrition due to servicing transfers

Transferred loans are not random. In fact, delinquent bank-held loans are much more likely to leave LPS coverage than comparable securitized loans. If the authors simply dropped those loans, the portfolio effect could be biased.

The hazard model directly addresses this issue, and the main foreclosure result remains strong even after allowing for transfer as a competing event.

4.4 Geography, housing markets, and other omitted factors

The paper also checks whether the result is driven by regional conditions:

- MSA fixed effects are always included,
- pooled regressions with zip-code fixed effects give similar estimates,
- controlling for house-price movements at the MSA level does not overturn the results.

Interpretation. The portfolio effect is not just a story about some banks holding loans in weaker neighborhoods or worse local markets.

4.5 Second liens and measurement concerns

Another concern is missing information about second liens. If securitized loans had more unobserved junior debt, they might naturally look harder to save. The authors proxy for this by adding an

$LTV = 80\%$ indicator, since that region is particularly likely to hide additional leverage. The result barely changes.

They also:

- change the delinquency threshold,
- broaden the foreclosure definition to include foreclosure starts,
- alter the definition of high-quality loans,
- and add quarter-of-delinquency fixed effects.

The qualitative conclusion survives all of these variants.

4.6 Why the quasi-experiment is persuasive

The repurchase-clause design is the paper's strongest evidence because it changes the *securitization status of an already delinquent loan*. That sharply reduces the importance of origination-stage unobservables.

The design is credible because:

- treatment and control loans near the 90-day cutoff look very similar on FICO, LTV, and interest rates,
- kernel densities and residualized scatterplots show no discontinuity in observables at the threshold,
- there is no bunching in delinquency incidence around the 90-day cutoff,
- a placebo comparison of month-3 loans that stay securitized versus month-4 loans that stay securitized shows no similar drop in foreclosure.

Interpretation. Once the balance and placebo tests work, the large drop in foreclosure for repurchased loans is hard to explain without a causal servicing effect.

4.7 What remains unresolved

The paper is careful about its limits:

- It does not fully distinguish between agency problems, legal frictions, coordination frictions, and accounting incentives.
- It does not fully settle whether banks are always closer to the efficient ex post benchmark.
- The quasi-experiment identifies a local treatment effect near the repurchase threshold, not necessarily the exact average effect for every delinquent loan in the economy.

Still, the evidence strongly supports the claim that securitization changes foreclosure behavior in economically meaningful ways.

5 Tables and figures

5.1 Table 1 and Table 2: who is delinquent, and how portfolio and securitized loans differ

Table 1
Summary statistics of all loans.

The sample only includes first lien loans. The investor is either private (securitized) or portfolio (bank balance sheet) at the time of the first observed month of 60+ days delinquency. Delinquent is defined as 60+ days MBA delinquent. Default is defined as a loan that enters into foreclosure post-sale or REO status. Age at delinquency is the number of months since origination when a loan becomes 60+ days delinquent. All loans in the sample are originated between 2005 and 2006.

Panel A: Delinquent loans

Origination quarter	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
% Portfolio	13.8%	12.5%	13.5%	10.7%	8.9%	8.6%	10.4%	11.7%
Original credit score	628.0	630.9	639.8	638.0	637.6	636.6	634.4	632.8
LTV	80.1	80.3	79.8	79.1	79.6	80.0	79.9	80.5
Original interest rate	7.02%	7.13%	7.13%	7.56%	8.08%	8.26%	8.45%	8.29%
Original loan amount	217,526	231,752	252,690	254,366	251,435	256,711	261,184	272,667
Age at delinquency	17.5	16.9	16.9	15.4	13.4	12.0	10.6	9.17
% Default/Foreclosure	24.19%	23.52%	22.73%	24.70%	26.27%	22.29%	19.93%	16.18%
N	35,585	46,521	46,907	45,133	42,978	42,354	37,386	30,574

Panel B: Delinquent loans by investor status

Portfolio	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Original credit score	639.2	656.2	656.3	662.9	664.7	660.1	634.0	641.6
LTV	78.9	79.2	79.4	79.1	80.3	81.3	82.2	83.2
Original interest rate	6.16%	6.29%	6.50%	6.67%	6.97%	7.54%	7.97%	7.64%
Original loan amount	248,033	282,570	271,062	305,099	297,276	286,659	249,147	264,680
Age at delinquency	17.4	16.9	14.9	14.2	12.8	11.0	9.0	8.0
% Default/Foreclosure	19.22%	19.26%	18.80%	20.00%	22.63%	19.18%	16.01%	15.35%
N	4,921	5,837	6,313	4,811	3,822	3,654	3,892	3,570

Securitized	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Original credit score	626.2	627.2	637.2	635.0	634.9	634.4	634.4	631.6
LTV	80.3	80.5	79.9	79.1	79.5	79.9	79.7	80.2
Original interest rate	7.15%	7.26%	7.23%	7.67%	8.19%	8.32%	8.50%	8.37%
Original loan amount	212,631	224,461	249,833	248,313	246,960	253,884	262,583	273,723
Age at delinquency	17.5	16.9	17.2	15.6	13.5	12.1	10.8	9.3
% Default/Foreclosure	24.99%	24.14%	23.35%	25.27%	26.62%	22.58%	20.38%	16.29%
N	30,664	40,684	40,594	40,322	39,156	38,700	33,494	27,004

Table 2
Summary statistics of high-quality loans (full documentation and FICO of at least 680).

The sample only includes first lien loans. The investor is either private (securitized) or portfolio (bank balance sheet) at the time of the first observed month of 60+ days delinquency. Delinquent is defined as 60+ days MBA delinquent. Default is defined as a loan that enters into foreclosure post-sale or REO status. Age at delinquency is the number of months since origination when a loan becomes 60+ days delinquent. All loans in the sample are originated between 2005 and 2006.

Panel A: Delinquent loans

Origination quarter	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
% Portfolio	17.6%	20.8%	21.5%	19.3%	20.7%	21.2%	17.8%	24.2%
Original credit score	716.5	718.3	718.8	718.5	717.6	716.2	715.6	717.8
LTV	79.9	80.2	79.6	78.7	78.4	79.1	79.0	79.4
Original interest rate	6.09%	6.29%	6.12%	6.53%	6.86%	7.16%	7.31%	7.19%
Original loan amount	250,483	256,730	280,300	276,557	276,597	297,623	311,906	320,919
Age at delinquency	21.2	20.0	19.4	17.8	15.8	13.5	11.7	9.8
% Default/Foreclosure	25.45%	25.08%	20.02%	21.01%	23.48%	20.44%	16.95%	13.67%
N	2,008	2,911	2,452	2,228	1,793	2,099	1,793	1,207

Panel B: Delinquent loans by investor status

Portfolio	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Original credit score	723.2	726.0	727.6	728.2	721.9	722.4	719.9	722.0
LTV	80.5	80.5	80.5	79.4	78.5	80.1	81.8	79.5
Original interest rate	5.13%	5.66%	5.44%	6.18%	6.54%	6.76%	6.89%	6.65%
Original loan amount	257,893	266,009	292,939	290,574	273,631	294,194	305,043	342,780
Age at delinquency	21.1	20.6	18.5	15.9	14.7	12.3	10.7	9.4
% Default/Foreclosure	19.26%	19.64%	14.61%	14.22%	18.28%	13.03%	8.44%	10.96%
N	353	606	527	429	372	445	320	292

Securitized	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Original credit score	715.1	716.3	716.4	716.2	716.4	714.6	714.7	716.5
LTV	79.8	80.2	79.4	78.5	78.4	78.9	78.3	79.4
Original interest rate	6.29%	6.46%	6.31%	6.62%	6.94%	7.27%	7.40%	7.36%
Original loan amount	248,903	254,290	276,839	273,214	277,374	298,546	313,397	313,942
Age at delinquency	21.2	19.8	19.6	18.3	16.2	13.8	11.9	10.0
% Default/Foreclosure	26.77%	26.51%	21.51%	22.62%	24.84%	22.43%	18.81%	14.54%
N	1,655	2,305	1,925	1,799	1,421	1,654	1,473	915

Read:

- Table 1 describes all delinquent loans. Across origination quarters, portfolio loans have higher FICO scores and lower interest rates than securitized loans, but they also have materially lower foreclosure rates.
- Table 2 repeats the exercise for high-quality loans only. In that sample, portfolio shares are larger and the ex post foreclosure gap is still present.
- The average delinquent loan takes roughly one to one-and-a-half years to become seriously delinquent, which is why updated borrower information at delinquency is potentially useful.

Economic message. The raw data already suggest that ownership status matters at the workout stage, but they also show why multivariate controls are necessary: portfolio and securitized loans are not identical on observables.

5.2 Table 3 and Table 4: baseline foreclosure regressions

Read:

- Table 3 is the central baseline result for all loans. The coefficient on *Portfolio* is negative in every origination quarter, ranging roughly from –4.6 to –7.0 percentage points.
- Relative to the mean foreclosure rate of delinquent securitized loans, that corresponds to about 18% to 32% lower foreclosure for comparable bank-held loans.
- Table 4 repeats the regression for the full-documentation, FICO $\geq$ 680 sample. The portfolio effect is generally even larger, reaching around –9.6 percentage points in some vintages, though the earliest quarter is estimated less precisely.

Economic message. The portfolio effect is not a one-off. It is pervasive across vintages, and it becomes stronger for borrowers who look better on hard information.

Table 3
Logit regression of default conditional on 60+ days delinquency (all loans).

This table reports the marginal effects of a logit regression. Coefficients on discrete variables represent the effect of moving from zero to one. Coefficients on continuous variables represent the effect of moving one standard deviation from the mean. Portfolio is a dummy which indicates that the loan was bank-held at the time of first 60+ days delinquency. Age at delinquency is the age of the loan at the time of first 60+ days delinquency. The excluded variables are private investor, FICO > =680, 30-year term, and missing insurance information. Standard errors are clustered at MSA level and resulting *t*-statistics are reported in parentheses. All loans in the sample are originated between 2005 and 2006. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Origination quarter	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Dependent variable: Foreclosure Mean securitized	0.2499	0.2414	0.2335	0.2527	0.2662	0.2258	0.2038	0.1629
Portfolio (d)	−0.046***	−0.048***	−0.046***	−0.070***	−0.059***	−0.060***	−0.066***	−0.038***
	(−8.12)	(−8.86)	(−8.21)	(−10.91)	(−8.21)	(−12.99)	(−12.97)	(−14.25)
FICO < 620 (d)	−0.109***	−0.133***	−0.127***	−0.145***	−0.155***	−0.124***	−0.108***	−0.069***
	(−11.15)	(−18.42)	(−17.92)	(−23.61)	(−19.81)	(−15.24)	(−16.43)	(−12.51)
620 < =FICO < 680 (d)	−0.025***	−0.037***	−0.034***	−0.038***	−0.042***	−0.030***	−0.028***	−0.017***
	(−3.57)	(−8.01)	(−6.36)	(−8.01)	(−7.82)	(−6.41)	(−4.97)	(−4.37)
LTV	0.579***	0.280***	0.501***	0.535***	0.553***	0.401***	0.100***	0.055***
	(6.47)	(4.50)	(7.10)	(6.68)	(7.37)	(5.14)	(3.56)	(3.18)
LTV squared	−0.405***	−0.163***	−0.342***	−0.361***	−0.373***	−0.265***	−0.035	−0.015
	(−5.73)	(−3.24)	(−6.20)	(−5.53)	(−6.16)	(−4.17)	(−1.45)	(−1.04)
Origination amount	−0.003	0.000	−0.001	−0.003	0.001	0.007	0.003	0.009**
	(−0.47)	(0.08)	(−0.19)	(−0.84)	(0.09)	(1.08)	(0.62)	(2.21)
Origination amount squared	0.009	0.001	−0.002	−0.001	0.001	−0.011	−0.008	−0.016**
	(1.64)	(0.26)	(−0.28)	(−0.16)	(0.19)	(−1.52)	(−1.42)	(−2.12)
Original interest rate	0.015***	0.012***	0.020***	0.018***	0.021***	0.015***	0.013***	0.010***
	(6.71)	(5.40)	(9.89)	(8.89)	(8.44)	(9.01)	(9.04)	(8.13)
FIX (d)	−0.081***	−0.070***	−0.058***	−0.060***	−0.053***	−0.046***	−0.036***	−0.026***
	(−15.34)	(−12.52)	(−13.62)	(−13.93)	(−7.24)	(−10.27)	(−7.55)	(−6.97)
15-Year term (d)	0.013	−0.047**	−0.074***	−0.060***	−0.108***	−0.028	0.114***	0.072*
	(0.48)	(−2.21)	(−3.12)	(−2.69)	(−5.50)	(−1.06)	(3.59)	(1.94)
20-Year term (d)	0.022	−0.053	−0.073*	−0.074	−0.086	−0.104***	−0.046	−0.050***
	(0.35)	(−1.27)	(−1.88)	(−1.47)	(−1.32)	(−3.31)	(−0.87)	(−2.91)
No insurance (d)	−0.018***	−0.016***	−0.002	0.004	0.013**	0.024***	0.014**	−0.002
	(−3.53)	(−2.81)	(−0.37)	(0.64)	(2.32)	(4.42)	(2.23)	(−0.59)
Insurance (d)	−0.019	−0.011	−0.015	0.009	−0.005	−0.019	−0.013	−0.004
	(−1.55)	(−0.98)	(−1.40)	(0.64)	(−0.27)	(−1.06)	(−0.99)	(−0.38)
Age at delinquency	−0.085***	−0.096***	−0.109***	−0.135***	−0.163***	−0.136***	−0.127***	−0.097***
	(−13.51)	(−17.02)	(−26.89)	(−32.76)	(−44.74)	(−51.75)	(−60.27)	(−126.99)
MSA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	35,365	46,279	46,636	44,904	42,789	42,050	37,008	29,939

Table 4
Logit regression of default conditional on 60+ days delinquency (high-quality loans).

This table reports the marginal effects of a logit regression. Coefficients on discrete variables represent the effect of moving from zero to one. Coefficients on continuous variables represent the effect of moving one standard deviation from the mean. Portfolio is a dummy which indicates that the loan was bank-held at the time of first 60+ days delinquency. Age at delinquency is the age of the loan at the time of first 60+ delinquency. The excluded variables are private investor, FICO > =760, 30-year term, and missing insurance information. Standard errors are clustered at MSA level and resulting *t*-statistics are reported in parentheses. All loans in the sample are originated between 2005 and 2006. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

Origination quarter	2005 Q1	2005 Q2	2005 Q3	2005 Q4	2006 Q1	2006 Q2	2006 Q3	2006 Q4
Dependent variable: Foreclosure Mean securitized	0.2677	0.2651	0.2151	0.2262	0.2484	0.2243	0.1881	0.1454
Portfolio (d)	−0.039	−0.057***	−0.041**	−0.066***	−0.079***	−0.096***	−0.089***	−0.045***
	(−1.29)	(−3.04)	(−1.98)	(−3.85)	(−2.88)	(−4.81)	(−7.20)	(−3.25)
680 < =FICO < 720 (d)	−0.028	0.002	0.050*	0.027	−0.023	0.037	−0.028	0.010
	(−0.87)	(0.06)	(1.89)	(1.20)	(−0.59)	(1.27)	(−1.15)	(0.35)
720 < =FICO < 760 (d)	−0.005	0.024	0.113**	0.026	−0.013	0.046	−0.028	0.010
	(−0.15)	(0.76)	(2.49)	(1.05)	(−0.32)	(1.25)	(−1.25)	(0.33)
LTV	0.529**	0.007	0.448***	0.213**	0.236**	0.351**	0.112*	−0.045
	(2.37)	(0.10)	(4.71)	(2.07)	(2.04)	(2.00)	(1.85)	(−1.48)
LTV squared	−0.439**	0.034	−0.355***	−0.143	−0.157	−0.281*	−0.067	0.058**
	(−2.19)	(0.55)	(−4.14)	(−1.56)	(−1.48)	(−1.74)	(−1.07)	(1.98)
Origination amount	−0.055	0.008	0.003	0.057	0.006	0.025	0.009	0.059**
	(−1.48)	(0.65)	(0.18)	(1.51)	(0.33)	(1.18)	(0.54)	(2.14)
Origination amount squared	0.164***	−0.012	0.001	−0.162*	0.002	−0.025	−0.010	−0.070*
	(2.63)	(−1.48)	(0.09)	(−1.73)	(0.10)	(−1.23)	(−0.64)	(−1.80)
Original interest rate	0.022	0.036***	0.026***	0.034***	0.007	0.040***	0.015*	0.015**
	(1.45)	(3.53)	(2.57)	(3.53)	(0.74)	(3.58)	(1.96)	(2.37)
FIX (d)	−0.085***	−0.108***	−0.049**	−0.053***	−0.078***	−0.006	−0.036**	0.001
	(−3.71)	(−6.34)	(−2.35)	(−2.83)	(−3.15)	(−0.38)	(−2.32)	(0.08)
15-Year term (d)	−0.040	−0.145***	−0.018	−0.103**	−0.067	−0.086	0.489**	0.040
	(−0.28)	(−3.03)	(−0.16)	(−2.31)	(−0.55)	(−1.23)	(2.12)	(0.43)
20-Year term (d)		0.041	−0.074	−0.032				
		(0.32)	(−0.85)	(−0.31)				
No insurance (d)	−0.014	−0.055**	−0.022	−0.041**	−0.006	0.011	−0.001	0.000
	(−0.58)	(−2.30)	(−1.22)	(−2.27)	(−0.25)	(0.39)	(−0.09)	(−0.01)
Insurance (d)	−0.040	−0.017	−0.024	−0.072***	0.091	−0.003	−0.028	−0.048**
	(−0.62)	(−0.45)	(−0.59)	(−2.89)	(0.93)	(−0.06)	(−0.62)	(−2.24)
Age at delinquency	−0.100***	−0.104***	−0.109***	−0.121***	−0.183***	−0.139***	−0.130***	−0.089***
	(−6.40)	(−9.09)	(−18.96)	(−16.17)	(−23.29)	(−23.59)	(−35.71)	(−30.00)
MSA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,758	2,631	2,123	1,978	1,555	1,826	1,518	905

5.3 Figure 1, Table 5, and Table 6: robustness to new information and sample attrition

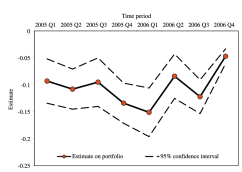

Fig. 1. Estimates on portfolio from logit regression using credit scores and LTV at time of delinquency. This figure reports the estimate (marginals) on the portfolio dummy using a specification similar to Table 3. We use FICO scores and LTV *at the time* of delinquency instead of using credit scores and LTV at the time of origination as in Table 3. Also plotted in the graph are the 95% confidence interval bands around the estimate.

Table 5
Hazard regression of default conditional on 60+ days delinquency (all loans and high-quality loans).

This table reports the estimated hazard ratios from a Cox-proportional hazard model of the transition from delinquency to foreclosure/transfer. Estimates on discrete variables represent the effect of moving from zero to one. Portfolio is a dummy which indicates that the loan was bank-held at the time of first 60+ days delinquency. All loans in the sample are originated between 2005 and 2006. Robust *t*-statistics are reported. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

	Panel A: All loans		*Panel B: High-quality loans*	
Dependent variable: Mean securitized	Foreclosure 0.25	Transfer 0.02	Foreclosure 0.25	Transfer 0.01
Portfolio (d)	0.759***	5.76***	0.662***	4.797***
	(−21.34)	(79.86)	(−8.28)	(12.17)
FICO	1.006***	1.000***	1.000	1.006***
	(91.77)	(−2.80)	(0.82)	(3.62)
LTV	1.162***	1.011*	1.113***	0.973
	(18.23)	(1.79)	(3.37)	(−1.27)
LTV squared	0.999***	1.000	0.999***	1.000
	(−16.18)	(−1.15)	(−2.80)	(1.48)
Origination amount	1.000***	1.000	1.000	1.001*
	(7.19)	(−0.09)	(−0.15)	(1.70)
Origination amount squared	1.000***	1.000***	1.000	1.000*
	(−4.41)	(−3.46)	(0.18)	(−1.74)
Original interest rate	1.162***	1.196***	1.110***	1.040
	(50.75)	(23.44)	(7.62)	(1.12)
FIX (d)	0.68***	0.82***	0.627***	1.66***
	(−39.57)	(−7.79)	(−11.53)	(4.09)
15-Year term (d)	0.778***	0.837	0.576*	1.318
	(−4.17)	(−1.60)	(−1.87)	(0.66)
20-Year term (d)	0.507***	0.485***	0.788	0.000***
	(−4.48)	(−2.74)	(−0.58)	(−143.93)
No insurance (d)	0.999	1.89***	0.852***	0.614***
	(−0.16)	(28.49)	(−4.10)	(−3.48)
Insurance (d)	0.903***	1.292***	0.734***	0.534**
	(−5.01)	(5.25)	(−3.26)	(−2.20)
Time fixed effects	Yes	Yes	Yes	Yes
MSA fixed effects	Yes	Yes	Yes	Yes
N	316,772	316,772	15,203	15,203

Table 6
Additional robustness tests.

This table reports the estimates (marginals) on Portfolio dummy using a specification similar to Table 3. The dependent variable is Foreclosure. The regression includes all the controls that were used in Table 3. Time and MSA (or zip-code) fixed effects are included in all specifications. High-quality loans consist of loans with FICO greater than or equal to 680 and full documentation. Coefficients on discrete variables represent the effect of moving from zero to one. All loans in the sample are originated between 2005 and 2006. More details on regressions in column (7) to (10) are provided in the internet appendix. Regressions in column (9) and (10) also include transferred loans. Standard errors are clustered at MSA level and resulting *t*-statistics are reported in parentheses. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

	Zip-code Fixed Effects		Alternative foreclosure definition		LTV=80 Dummy		Quarter of Delinquency Fixed Effects			
							Logit regressions		*Hazard regressions*	
	(1) All loans	(2) High-quality	(3) All loans	(4) High-quality	(5) All loans	(6) High-quality	(7) All loans	(8) High-quality	(9) All loans	(10) High-quality
Mean securitized	0.23	0.23	0.56	0.54	0.23	0.23	0.23	0.23	0.25	0.25
Portfolio (d)	−0.057***	−0.063***	−0.101***	−0.124***	−0.051***	−0.057***	−0.030***	−0.053***	0.772***	.715***
	−24.73	−4.03	−19.69	−9.96	−15.04	−6.57	(−14.13)	(−14.03)	(−19.55)	(−6.40)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter of origination fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter of delinquency fixed effects	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Clustering unit	MSA	MSA	MSA	MSA	MSA	MSA	MSA	MSA	MSA	MSA
Other fixed effects	Zip	Zip	MSA	MSA	MSA	MSA	MSA	MSA	MSA	MSA
N	327,438	16,491	327,372	16,272	327,401	16,106	327,438	13,822	316,772	15,203

Read:

- Figure 1 shows that when the authors replace origination FICO and origination LTV with updated values measured at delinquency, the estimated portfolio effect stays negative and statistically significant in every quarter.
- Table 5 uses hazard models that explicitly treat transfers out of LPS as a competing risk. Even though bank-held delinquent loans are much more likely to be transferred, they are still about 24% less likely to be foreclosed in the full sample and about 34% less likely in the high-quality sample.
- Table 6 shows that zip-code fixed effects, alternative foreclosure definitions, a proxy for hidden second liens, and quarter-of-delinquency fixed effects all preserve the main result.

Economic message. The basic result does not disappear once the authors add more information, more geography, or more careful treatment of missing outcomes.

5.4 Table 7 and Table 8: cure rates and heterogeneity by borrower quality

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Table 7
Hazard regression of cure rate conditional on 60+ days delinquency (all loans and high-quality loans).

This table reports the estimated hazard ratios from a Cox-proportional hazard model of the transition from delinquency to cure/transfer. A mortgage is considered cured if it makes another payment following delinquency. Estimates on discrete variables represent the effect of moving from zero to one. Portfolio is a dummy which indicates that the loan was bank-held at the time of first 60+ days delinquency. All loans in the sample are originated between 2005 and 2006. Robust *t*-statistics are reported. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

	Panel A: All loans		*Panel B: High-quality loans*	
Dependent variable:	Cure	Transfer	Cure	Transfer
Mean securitized	0.47	0.02	0.40	0.01
Portfolio (d)	1.129***	6.226***	1.205***	8.261***
	(17.15)	(71.91)	(6.69)	(14.31)
FICO	0.996***	1.00**	1.000	1.006***
	(−98.15)	(−2.08)	(−0.22)	(3.44)
LTV	0.991***	0.996	0.992	0.965*
	(−8.67)	(−0.57)	(−1.64)	(−1.71)
LTV squared	1.000***	1.000	1.000*	1.000
	(−7.01)	(0.54)	(−1.78)	(1.55)
Origination amount	1.000***	1.000**	1.000***	1.002**
	(−20.62)	(−2.18)	(−3.19)	(2.13)
Origination amount squared	1.000***	1.000*	1.000	1.000**
	(8.08)	(−1.81)	(1.28)	(−2.13)
Original interest rate	0.892***	1.166***	0.907***	1.046
	(−73.59)	(16.78)	(−12.52)	(1.23)
FIX (d)	1.227***	0.999	1.416***	2.655***
	(39.79)	(−0.02)	(14.44)	(6.92)
15-Year term (d)	1.125***	0.812	1.178**	1.828
	(6.10)	(−1.39)	(2.25)	(1.33)
20-Year term (d)	1.177***	0.458**	1.159	0.000***
	(4.22)	(−2.18)	(1.41)	(−127.51)
No insurance (d)	1.011**	2.114***	1.071***	0.406***
	(2.03)	(28.05)	(2.82)	(−5.49)
Insurance (d)	1.196***	1.259***	1.258***	0.259***
	(13.56)	(3.78)	(4.11)	(−3.51)
Time fixed effects	Yes	Yes	Yes	Yes
MSA fixed effects	Yes	Yes	Yes	Yes
N	316,772	316,772	15,203	15,203

Table 8
Hazard regression of default and cure conditional on 60+ days delinquency (cross-sectional evidence).

This table reports the estimated hazard ratios from Cox-proportional hazard models of the transition from delinquency to foreclosure/transfer and of the transition from delinquency to cure/transfer for different FICO segments. Estimates on discrete variables represent the effect of moving from zero to one. Portfolio is a dummy which indicates that the loan was bank-held at the time of first 60+ days delinquency. All loans in the sample are originated between 2005 and 2006. Robust *t*-statistics are reported. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

	$FICO < 620$		$620 < FICO < 680$		$FICO > 680$	
Dependent variable:	Foreclosure	Transfer	Foreclosure	Transfer	Foreclosure	Transfer
Mean securitized	0.21	0.02	0.28	0.02	0.28	0.02
Portfolio (d)	1.044*	9.859***	0.734***	5.39***	0.67***	1.915***
	(1.95)	(76.57)	(−14.24)	(44.33)	(−18.32)	(12.46)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
MSA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	131,466	131,466	111,638	111,638	73,668	73,668

	$FICO < 620$		$620 < FICO < 680$		$FICO > 680$	
Dependent variable:	Cure	Transfer	Cure	Transfer	Cure	Transfer
Mean securitized	0.58	0.02	0.41	0.02	0.34	0.02
Portfolio (d)	0.967***	9.462***	1.171***	6.35***	1.349***	2.829***
	(−3.01)	(61.16)	(12.53)	(4.79)	(22.53)	(17.76)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
MSA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	131,466	131,466	111,638	111,638	73,668	73,668

- Table 7 shows that delinquent bank-held loans are also more likely to cure. In the full sample, cure rates are about 6 percentage points higher within a year after delinquency; in the high-quality sample, the advantage is about 8.2 percentage points.
- Table 8 splits the sample by initial FICO. For FICO below 620, foreclosure and cure differences are economically small. For the 620–680 group, bank-held loans are foreclosed about 7.4 percentage points less and cured about 7.0 percentage points more. For FICO above 680, the foreclosure difference is about 9.2 percentage points and the cure difference about 12.1 percentage points.

Economic message. The paper's strongest effects come from borrowers for whom renegotiation is most plausibly valuable. That pattern is difficult to reconcile with a crude "bad loans get securitized" story.

5.5 Figure 2 and Table 9: the repurchase threshold and the balance of the quasi-experimental sample

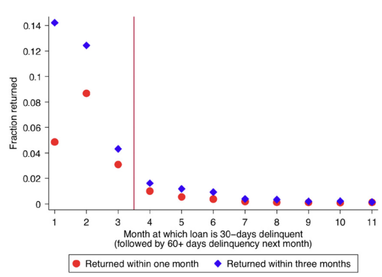

Fig. 2. Fraction of delinquent loans coming back to lenders. This figure plots the fraction of loans that return on portfolio as a function of the first time they become delinquent after securitization. More precisely, the figure shows a fraction of loans that becomes 30-day delinquent followed by 60-day delinquency for the first time in a given month after securitization that return on portfolio. We present the graph for two horizons–returns within one month after the loan hits 60-day delinquency and returns within three months after the loan hits 60-day delinquency. Consistent with the time horizon over which repurchase clauses are applicable (see Section 6), we observe that most returns for delinquent loans occur in the first three months after securitization.

Table 9
Summary statistics of sample of loans used in quasi-experiment.

This table reports summary statistics of a sample of loans used to conduct the test that exploits the repurchase clauses. The vertical headers indicate how many months after securitization the loan first becomes 30+ days delinquent, followed by 60+ days the next month. The top (bottom) panel tracks the loan for one (three) month(s) after the month of 60+ days delinquency. The treatment group consists of securitized loans that return to portfolio either in the month of 60+ days delinquency or within the corresponding time horizon. The control group consists of loans that remain securitized throughout the corresponding time horizon.

Panel A: One-month tracking horizon

Months after securitization delinquent	One month				Two months				Three months				Three vs. Four months			
	Treatment		Control		Treatment		Control		Treatment		Control		Treatment		Control	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
FICO	615.3	63.3	613.6	65.5	618.7	63.3	613.3	65.6	614.7	60	612.7	63.8	614.7	60	610.8	62.2
LTV	82.2	11.6	80.5	12.3	82.2	11.8	80.3	11.2	81.7	12.2	80.4	11.1	81.7	12.2	80.5	11.4
Interest rate	0.089	0.016	0.083	0.015	0.086	0.015	0.084	0.016	0.082	0.014	0.083	0.015	0.082	0.014	0.083	0.015
Foreclosure	0.46	0.5	0.36	0.48	0.41	0.49	0.39	0.49	0.31	0.46	0.37	0.48	0.31	0.46	0.35	0.48
N	390	390	7,610	7,610	1,041	1,041	10,849	10,849	394	394	12,345	12,345	394	394	12,824	12,824

Panel B: Three-month tracking horizon

Months after securitization delinquent	One month				Two months				Three months				Three vs. Four months			
	Treatment		Control		Treatment		Control		Treatment		Control		Treatment		Control	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
FICO	616.1	60.4	611.1	65.1	618.2	61.6	613.4	65.6	617.6	61.1	613.1	63.5	617.6	61.1	611.2	62.0
LTV	82.7	12.2	80.4	11.8	81.7	11.8	80.5	11.1	82.1	11.5	80.5	11.1	82.1	11.5	80.7	11.2
Interest rate	0.087	0.014	0.084	0.015	0.085	0.015	0.084	0.016	0.082	0.015	0.083	0.015	0.082	0.015	0.083	0.015
Foreclosure	0.41	0.49	0.4	0.49	0.4	0.49	0.41	0.49	0.32	0.47	0.39	0.49	0.32	0.47	0.36	0.48
N	1,044	1,044	6,240	6,240	1,489	1,489	9,934	9,934	526	526	11,636	11,636	526	526	12,157	12,157

Read:

- Figure 2 shows a sharp institutional pattern: about 9% of securitized loans that become delinquent within the first three months after securitization are repurchased by the lender within the next three months, while only about 1% of comparable loans delinquent after month three come back.

- Table 9 then compares treatment and control loans around the 90-day threshold. The groups look similar on key observables. For example, FICO is about 617 in the treatment group versus 611 in the control group, the interest rate is about 8.2% versus 8.3%, and LTV is about 82.1 versus 80.7.

Economic message. The institutional discontinuity is real, and the loans just around the cutoff are sufficiently comparable to make the quasi-experiment credible.

5.6 Figure 3, Figure 4, Figure 5, and Table 10: causal evidence from the quasi-experiment

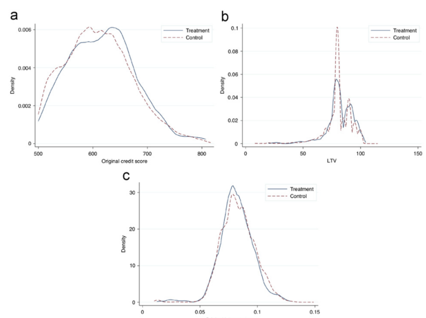

Fig. 3. Kernel densities of characteristics of loans in treatment and control group used in the main (3:4) test in the quasi-experiment. This figure shows Epanechnikov kernel densities of characteristics of the loan sample used to conduct the test that exploits the repurchase clauses. The treatment group consist of securitized loans which became 30+ days delinquent three months after securitization, transitioned to 60+ in the next month, and were repurchased by the originator in the following three months, while the control group consist of securitized loans which became 30+ days delinquent four months after securitization, transitioned to 60+ in the next month, and remained securitized in the next three months. The bandwidth employed has the width that minimizes the mean integrated squared error as if the data were Gaussian and a Gaussian kernel were used.

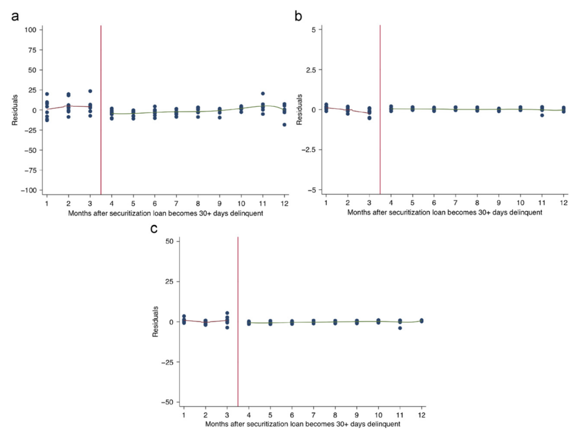

Fig. 4. Scatter plot of loan characteristics around 90-day threshold. This figure shows the scatter plot of characteristics of loans around the 90-day threshold. To the left of the cutoff, we represent average characteristics of loans in every quarter which became 30+ days delinquent in a given month, transitioned to 60+ in the next month, and were repurchased by the originator in the subsequent month (tracked for three months after 60+). To the right of the cutoff, we represent characteristics of loans in every quarter which became 30+ days delinquent in a given month, transitioned to 60+ in the next month, and remained securitized in the subsequent month (tracked for three months after 60+). By construction, characteristics of loans in month three and month four in the graph represent the characteristics of loans in the treatment and control group of the 3:4 test. To remove any time effects, we regress loan characteristics on quarter dummies and plot the residuals. Figure (a) plots FICO scores, Figure (b) plots interest rates, while Figure (c) plots LTV ratios.

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- Figure 3 plots kernel densities for treatment and control loans and shows that the distributions of key observables line up closely.

- Figure 4 shows no jump in FICO, interest rate, or LTV around the 90-day cutoff after residualizing out time effects.

- Figure 5 shows a visible drop in foreclosure rates for loans on the treatment side of the threshold.

- Table 10 formalizes the result. In the main 3:4 specification, repurchased loans are foreclosed about 6.5 percentage points less often than comparable loans that remain securitized.

- The placebo version of the 3:4 test – replacing the treatment group with month-3 loans that remain securitized – yields a small, statistically insignificant estimate.

- The 3:3 comparison gives an effect of about –6.9 percentage points, and pooled versions of the design continue to show economically meaningful declines in foreclosure.

Economic message. This is the paper's cleanest evidence that the foreclosure gap is causal. Once the same type of distressed loan is pushed back onto a lender's balance sheet, foreclosure drops sharply.

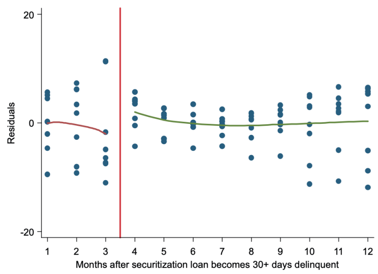

Fig. 5. Scatter plot of foreclosure rates around 90-day threshold. This figure shows the scatter plot of foreclosure rate of loans around the 90-day threshold. To the left of the cutoff, we represent average foreclosure rate of loans in every quarter which became 30+ days delinquent in a given month, transitioned to 60+ in the next month, and were repurchased by the originator in the subsequent month (tracked for three months after 60+). To the right of the cutoff, we represent foreclosure rate of loans in every quarter which became 30+ delinquent in a given month, transitioned to 60+ in the next month, and remained securitized in the subsequent month (tracked for three months after 60+). By construction, foreclosure rates of loans in month three and month four in the graph represent the foreclosure rate of loans in the treatment and control groups of the 3:4 test. To remove any time effects, we regress foreclosure rates on origination quarter dummies. The regression also conditions on other observables such as FICO, LTV, origination amount and interest rates and plot the residuals.

Table 10
Regression estimates using a quasi-experiment.

This table reports the estimates (marginals) using a specification and controls as used in Table 3. The dependent variable is Foreclosure. In columns (1) and (2), the treatment group consist of securitized loans which became 30+ days delinquent three months after securitization, transitioned to 60+ in the next month, and were repurchased by the originator in the following three months, while the control group consist of securitized loans which became 30+ days delinquent four months after securitization, transitioned to 60+ in the next month, and remained securitized in the next three months. In column (3), the treatment group consists of securitized loans which became 30+ days delinquent three months after securitization, transitioned to 60+ in the next month, and remained securitized in the next three months, while the control group consists of securitized loans which became 30+ days delinquent four months after securitization, transitioned to 60+ in the next month, and remained securitized in the next three months. In column (4), the treatment group consist of securitized loans which became 30+ days delinquent three months after securitization, transitioned to 60+ in the next month, and were repurchased by the originator in the following three months, while the control group consist of securitized loans which became 30+ days delinquent three months after securitization, transitioned to 60+ in the next month, and remained securitized in the next three months. In column (5), the treatment group consists of securitized loans which became 30+ days delinquent within three months after securitization, transitioned to 60+ in the next month, and were repurchased by the originator in the following three months, while the control group consists of securitized loans which became 30+ days delinquent within three months after securitization, transitioned to 60+ in the next month, and remain securitized in the three months following delinquency. In column (6), the treatment group consists of securitized loans which became 30+ days delinquent within three months after securitization, transitioned to 60+ in the next month, and were repurchased by the originator in the following three months, while the control group consists of securitized loans which became 30+ days delinquent between four and six months after securitization, transitioned to 60+ in the next month, and remain securitized in the three months following delinquency. In column (7), the treatment group and the control group are same as in column (5). In addition to the Portfolio dummy, the regression includes FICO, LTV, interest rate, origination amount, squared terms of these variables, insurance and maturity dummies, age of the loan at delinquency, and MSA and origination-quarter fixed effects. Marginal effects are reported for the logit regression. Coefficients on discrete variables represent the effect of moving from zero to one. Standard errors are clustered at MSA level and resulting *t*-statistics are reported in parentheses. ***, **, and * represent significance at 1%, 5%, and 10%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Test (3-month tracking horizon)	3:4 (Main)	3:4 (Main)	3:4 (Placebo)	3:3 (Main)	1:1+2:2+3:3 (Pooled)	123:456 (Pooled)	1:1+2:2+3:3 (Pooled)
Mean control	0.364	0.364	0.364	0.391	0.401	0.367	0.401
Portfolio (d)	−0.049*	−0.065**	0.011	−0.069**	−0.041**	−0.033**	−0.031*
	(−1.80)	(−2.49)	(1.39)	(−2.44)	(−2.32)	(−1.97)	(1.93)
Portfolio (d)*High-quality (d)							−0.115***
							(−2.63)
High-quality (d)							−0.028**
							(−2.30)
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MSA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loan age	No	Yes	Yes	Yes	Yes	Yes	Yes
Loan origination period	05Q1–06Q4	05Q1–06Q4	05Q1–06Q4	05Q1–06Q4	05Q1–06Q4	05Q1–06 Q4	05Q1–06Q4
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustering unit	MSA	MSA	MSA	MSA	MSA	MSA	MSA
Treatment/Control	481/11,196	481/11,196	10,816/11,313	474/10,672	2,880/25,913	2,887/59,211	2,880/25,913

5.7 Table 11 and Figure 6: why the threshold design is not a manipulation artifact

Table 11
Characteristics of EPD and non-EPD deals.

Panel A presents summary statistics for loans originated in securitization pools with and without early payment default clauses. Panel B reports the estimates (marginals) of a logit regression. The dependent variable is 60+ days delinquency in the first two columns and 30+ delinquency in the last two columns. Control variables include FICO score, FICO score squared, LTV, interest rate, loan amount, MSA fixed effects, and origination time fixed effects where indicated. Standard errors are clustered at MSA level and resulting t-statistics are reported in parentheses. ***, **, and * represents significance at 1%, 5%, and 10% respectively. Data comes from the BlackBox loan database and spans 2005 and 2006.

Panel A: Summary statistics

	EPD Deals	Non-EPD Deals	Difference
FICO	616.8	645.7	28.93***
LTV	78.7	78.6	−.1
Interest Rate	8.37	7.43	−.94***
Observations	99,809	484,398	

Panel B: Logit regressions

	Pr (60+ =1)	Pr (60+ =1)	Pr (30+ = 1)	Pr (30+ = 1)
EPD dummy	0.009	−0.015	−0.001	−0.023
	(0.33)	(−0.57)	(−0.03)	(−0.83)
FICO	−0.138	−0.157*	0.017	−0.001
	(−1.41)	(−1.67)	(0.20)	(−0.01)
FICO squared	−0.017	−0.002	−0.208**	−0.194**
	(−0.18)	(−0.02)	(−2.42)	(−2.32)
LTV	−0.112**	−0.125**	−0.090*	−0.103**
	(−1.97)	(−2.23)	(−1.79)	(−2.09)
Interest rate	0.208***	0.160***	0.042***	0.035**
	(3.49)	(3.01)	(2.91)	(2.51)
Other controls	Yes	Yes	Yes	Yes
Observations	584,183	584,183	584,204	584,204
Time fixed effects	No	Yes	No	Yes
MSA fixed effects	Yes	Yes	No	Yes

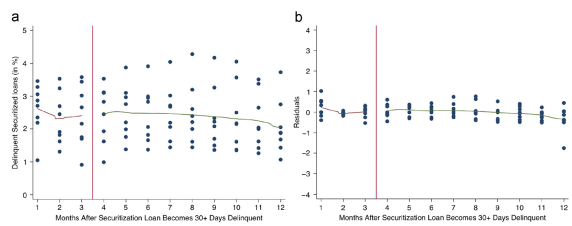

Fig. 6. Scatter plot of percentage of loans becoming 30+ days delinquent around 90-day threshold. This figure shows the scatter plot of percentage of securitized loans in every quarter that become 30+ days delinquent around the 90-day threshold. We present both the raw percentages (a) and residuals obtained after regressing the raw percentages on origination quarter dummies to remove the time effects (b).

Read:

- Table 11 compares securitization deals with and without explicit early payment default clauses. Loans in EPD deals actually look *riskier* on observables: average FICO is about 617 versus 646 and average interest rate is about 8.37% versus 7.43%.
- Conditional on observables, those loans do not display abnormal delinquency rates that would suggest a hidden selection toward safer loans in EPD deals.
- Figure 6 shows that the fraction of securitized loans becoming 30+ delinquent is smooth around the 90-day threshold. There is no sign that lenders systematically hold bad loans just long enough to slip them past the repurchase cutoff.

Economic message. The threshold design is not obviously driven by safer loans being placed into EPD deals or by strategic bunching of delinquency just after the cutoff.

6 Short summary

The paper shows that securitization materially changes how seriously delinquent mortgages are handled. Using a very large loan-level servicing dataset, the authors find that delinquent bank-held loans are foreclosed substantially less often than comparable securitized loans. The result survives rich controls, updated information at delinquency, hazard models that account for servicing transfers, and extensive robustness checks. It is especially strong for medium- and high-quality borrowers. A quasi-experiment based on repurchase clauses around a 90-day securitization threshold provides causal evidence: when distressed loans are pushed back onto the originator's balance sheet, foreclosure rates fall sharply. The broad conclusion is that securitization introduced renegotiation frictions that worsened foreclosure outcomes during the subprime crisis.

7 Conclusion

7.1 Core conclusion

The paper establishes that ownership structure matters after a mortgage becomes distressed. Securitized delinquent loans are not serviced the same way as loans kept on the lender's balance sheet, and the difference is large enough to affect aggregate foreclosure outcomes in a crisis.

7.2 Economic intuition

A lender that owns the loan internalizes the full cost and benefit of foreclosure versus workout. Once the loan is securitized, servicers may face weaker incentives, legal ambiguity, or coordination frictions among investors. Those frictions do not need to prevent every renegotiation; they only need to tilt the margin toward foreclosure often enough to matter.

7.3 What the heterogeneity tells us

The fact that the largest differences show up for better-quality borrowers is especially informative. Those are precisely the borrowers for whom a successful workout is more plausible, because the chance of re-default is lower. The heterogeneity therefore points toward real renegotiation frictions rather than simple compositional differences.

7.4 Policy takeaway

The paper does not prove that banks always choose the socially optimal outcome, but it does show that securitization can intensify foreclosure during systemic stress. Since foreclosure creates neighborhood spillovers and can further depress collateral values, the result provides a strong rationale for policies that facilitate renegotiation when large shocks hit.

7.5 Bottom line

The big lesson is that securitization changed more than mortgage funding. It changed the control rights and incentives governing distress, and those contractual differences helped shape the foreclosure crisis.